SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

# Shoulder instability

## 1. Anatomy and stabilisers

- **Glenohumeral joint:** the most mobile and most commonly dislocated large joint. **Glenoid surface area is about one-third to one-quarter of the humeral head** ("golf ball on a tee"); glenoid **retroverted about 5-7 degrees**, superiorly tilted about 5 degrees; humeral head retroverted about 30 degrees.
- **Static stabilisers:**
  - **Labrum** (deepens the glenoid by about **50%**, increases contact area, anchors the IGHL; **sublabral foramen** and **Buford complex (absent anterosuperior labrum + cord-like MGHL)** are **normal variants**: do not repair).
  - **Glenohumeral ligaments:**
  - **SGHL:** restraint to **inferior translation in adduction** (with the coracohumeral ligament; **rotator interval**),
  - **MGHL: anterior translation at about 45 degrees abduction** with external rotation (absent in about 30%),
  - **IGHL complex (anterior band, axillary pouch, posterior band): the most important**: **anterior band** resists **anterior translation in 90 degrees abduction and external rotation** (the **apprehension position**); **posterior band** resists **posterior translation in flexion and internal rotation**.
  - **Coracohumeral ligament**: inferior translation in adduction, external rotation (tight in frozen shoulder).
  - **Negative intra-articular pressure**, **concavity-compression**, **adhesion-cohesion**.
- **Dynamic stabilisers: rotator cuff (concavity-compression: most important dynamic)**, **long head of biceps**, **scapular stabilisers**

(scapulothoracic rhythm about **2:1 glenohumeral:scapulothoracic**).

- **Rotator interval:** triangle between **supraspinatus and subscapularis** (base: coracoid): contains **SGHL, CHL, long head of biceps**: **interval closure** reduces inferior/posterior laxity (MDI).

## 2. Classification of instability

- **TUBS** (**T**raumatic, **U**nidirectional, **B**ankart lesion, **S**urgery) vs **AMBRI** (**A**traumatic, **M**ultidirectional, **B**ilateral, **R**ehabilitation, **I**nferior capsular shift if surgery). **Stanmore triangle (polar types):**
  - **Polar I:** traumatic structural,
  - **Polar II:** atraumatic structural (capsular laxity/dysplasia),
  - **Polar III: muscle patterning (abnormal muscle recruitment)**, non-structural: **physiotherapy/biofeedback; avoid surgery**.
- **Direction: anterior (about 95%)**, posterior (**about 2-5%: seizures, electrocution, posterior force with adduction/internal rotation**), inferior (**luxatio erecta**: arm fixed overhead; **axillary nerve and artery injuries**), multidirectional.

## 3. Traumatic anterior dislocation

- **Mechanism: abduction + external rotation + extension force**.
- **Associated lesions:**
  - **Bankart lesion** (**anteroinferior labrum + IGHL avulsion from the glenoid**): **in about 90% of young patients**. **Bony Bankart** (glenoid rim fracture).
  - **Hill-Sachs lesion** (**posterosuperolateral humeral head impaction** on the glenoid rim: **in about 80-90%**).
  - **ALPSA** (**anterior labroligamentous periosteal sleeve avulsion**: labrum displaced **medially and inferiorly** on the scapular neck: heals in a malposition: **higher recurrence**; mobilise before repair).

- **Perthes lesion** (labrum avulsed with **intact periosteum**: non-displaced).
- **GLAD** (**glenolabral articular disruption**: labral tear + articular cartilage injury; usually stable).
- **HAGL** (**humeral avulsion of the glenohumeral ligament**: **"J sign"** on MRI (IGHL becomes J-shaped); **recurrent instability if missed**; **open or arthroscopic repair**).
- **Greater tuberosity fracture** (**commoner in over 40s**; reduces with the shoulder in most cases; **displacement over 5 mm (over 3 mm in overhead athletes): fix**).
- **Rotator cuff tears** (**over 40 years: about 30%; over 60 years: about 80%**: **"terrible triad" of the shoulder**: dislocation + cuff tear + nerve injury). **Persistent weakness after reduction in an older patient = cuff tear (until proved otherwise)**: ultrasound/MRI; **early repair**.
- **Axillary nerve injury** (**about 5-35%, mostly neurapraxia**; **older patients**): check **regimental badge sensation** and deltoid **before and after reduction**.
- **Axillary artery injury** (elderly with atherosclerosis).

- **Reduction techniques: Kocher** (traction, external rotation, adduction, internal rotation: **risk of fracture/nerve injury**), **Hippocratic**, **Stimson** (prone, hanging weight), **Milch** (abduction + external rotation), **FARES**, **Cunningham (massage technique)**, **scapular manipulation**; **post-reduction radiographs (AP + axillary/Velpeau)**.
- **Recurrence (the most important determinant is age at first dislocation):**
  - **Under 20 years: about 70-90%** (up to about **90%** in **contact athletes**, male),
  - **20-40 years: about 50-60%**,

  - **Over 40 years: about 10-15%** (cuff tears, stiffness more common than recurrence).
- **Immobilisation in external rotation** (to approximate the Bankart lesion) has **not consistently reduced recurrence** vs internal rotation sling.
- **Primary arthroscopic stabilisation after first dislocation** in **young (under about 25) contact/collision athletes** reduces recurrence (**about 80% recurrence non-operatively vs about 10-15% after surgery**); offered selectively.

## 4. Bone loss: the key to surgical choice

### Glenoid bone loss

- **Inverted pear glenoid** (arthroscopic appearance: loss of anteroinferior bone, **over about 25% loss**).
- **Measurement: 3D CT (en face view of the glenoid)**: **best-fit inferior circle method**, **Pico method**, **glenoid width loss as percentage**.
- **Critical bone loss: historically about 20-25%**; **"subcritical" bone loss of about 13.5-17%** still raises recurrence and worsens outcomes after arthroscopic Bankart repair (**Shaha**).

### Humeral bone loss and the glenoid track

- **Glenoid track concept (Yamamoto, Di Giacomo):** the zone of contact of the glenoid on the humeral head in **abduction/external rotation**: **glenoid track width = 0.83 x D (intact glenoid width) - d (glenoid bone loss width)**.
  - **Hill-Sachs interval (HSI) = width of the Hill-Sachs lesion + bone bridge** (to the cuff insertion).
  - **HSI over the glenoid track = "off-track" (engaging) Hill-Sachs** (it can **lever over the anterior glenoid rim** in the functional range): **high recurrence after isolated Bankart repair**.

  - **HSI under the glenoid track = "on-track"**.
- **Engaging Hill-Sachs (Burkhart):** lesion aligned parallel to the anterior glenoid in a functional position (abduction/external rotation): engages.

### Treatment algorithm (Di Giacomo et al.)

| Glenoid bone loss | Hill-Sachs | Treatment |
|---|---|---|
| **Under 25% (or under subcritical threshold)** | **On-track** | **Arthroscopic Bankart repair** |
| **Under 25%** | **Off-track** | **Arthroscopic Bankart + remplissage** |
| **Over 25%** | On-track | **Latarjet (bone block)** |
| **Over 25%** | Off-track | **Latarjet ± humeral procedure (remplissage/bone graft)** depending on track after Latarjet |

- **ISIS score (Instability Severity Index Score; Balg and Boileau): over 6 points = high recurrence after arthroscopic Bankart (about 70%): Latarjet**:
  - **Age under 20 at surgery: 2 points**,
  - **Competitive sport: 2 points**,
  - **Contact or forced overhead sport: 1 point**,
  - **Shoulder hyperlaxity (external rotation over 85 degrees or Gagey hyperabduction over 90 degrees): 1 point**,
  - **Hill-Sachs visible on AP radiograph in external rotation: 2 points**,
  - **Loss of glenoid contour on AP radiograph: 2 points**.

## 5. Surgical procedures

### Arthroscopic Bankart repair

- **Suture anchors (at least 3 anchors, at least 4 mm apart, placed on the articular margin of the glenoid face (not medially on the neck))**, **inferior capsular shift** (plication), **anchors at the 3-5 o'clock (right shoulder)** region; **beach chair or lateral decubitus**.
- **Recurrence about 5-15% (higher with bone loss, young contact athletes, ISIS over 6, fewer than 3 anchors, ALPSA)**.
- **Open Bankart** (subscapularis split/takedown): historically the gold standard; **contact athletes** in some centres; **external rotation loss**.

### Remplissage ("to fill")

- **Arthroscopic capsulotenodesis of the infraspinatus tendon (and posterior capsule) into the Hill-Sachs defect**: converts the intra-articular defect to extra-articular. **Small loss of external rotation**; reduces recurrence for off-track lesions.

### Latarjet (coracoid transfer)

- **Coracoid process (about 2-3 cm) with the conjoint tendon** transferred to the **anteroinferior glenoid (flush with the glenoid face, 3-5 o'clock, below the equator)** through a **subscapularis split**, fixed with **two screws**.
- **"Triple blocking effect" (Patte):**
  - **Bony effect** (extends the glenoid arc),
  - **Sling effect of the conjoint tendon** (**most important in abduction-external rotation**: dynamic sling on the lower subscapularis and capsule),
  - **Capsular repair** (to the coracoacromial ligament stump: **Bankart effect**).
- **Bristow** = tip of the coracoid only, placed vertically, fixed with one screw (less bone).

- **Complications: graft malposition (too lateral: OA; too medial: recurrence)**, **graft non-union/osteolysis/fracture**, **screw problems**, **musculocutaneous nerve injury** (enters the coracobrachialis about **5 cm below the coracoid**: beware traction), **axillary nerve injury**, **loss of external rotation**, **glenohumeral OA (long-term)**, recurrence (low, about 2-5%), infection, haematoma.
- **Free bone block (iliac crest (Eden-Hybinette), distal tibial allograft)** for **massive bone loss or failed Latarjet**.

### Other

- **HAGL repair**, **humeral head allograft/disimpaction** for large Hill-Sachs, **humeral rotational osteotomy (historical: Weber)**.
- **Thermal capsulorrhaphy: abandoned** (capsular necrosis, **chondrolysis**, axillary nerve injury). **Intra-articular bupivacaine pain pumps**: **glenohumeral chondrolysis**.

## 6. Posterior instability

- **Traumatic posterior dislocation: seizures, electrocution, ECT**: **often missed (about 50-80% missed initially)**; **arm locked in internal rotation**, **cannot externally rotate**; **AP radiograph: "light bulb sign"**, loss of the half-moon overlap, **rim sign (over 6 mm)**; **axillary view/CT essential**. **Reverse Hill-Sachs lesion (McLaughlin lesion)** of the **anteromedial humeral head**.
  - **Treatment by defect size: under 25%**: closed reduction, immobilise in external rotation; **25-50%**: **McLaughlin procedure (subscapularis tendon transfer into the defect)** or **modified McLaughlin (lesser tuberosity transfer: Neer)**, **allograft reconstruction**; **over 50%** or chronic over 6 months: **hemiarthroplasty/arthroplasty**.
- **Recurrent posterior subluxation: athletes (football linemen, weightlifters, swimmers)**: **reverse Bankart lesion**, **Kim lesion** (incomplete, concealed avulsion of the posteroinferior labrum); **Jerk**

**test, Kim test**; **posterior labral repair/capsular shift**; **posterior bone block** for glenoid bone loss; **glenoid opening-wedge osteotomy** for **excess glenoid retroversion** (over about 15-20 degrees).

## 7. Multidirectional instability (MDI)

- **Inferior + anterior and/or posterior instability**; **generalised hyperlaxity (Beighton score)**, connective tissue disorders (**Ehlers-Danlos**), **patulous inferior capsule**, enlarged rotator interval; often **bilateral, atraumatic**.
- **Sulcus sign (over 2 cm, persistent in external rotation = rotator interval incompetence)**, **Gagey hyperabduction test** (**over 105 degrees = IGHL laxity**).
- **Treatment: prolonged rehabilitation (at least 6 months)**: rotator cuff and scapular stabiliser strengthening (**most succeed**); **surgery** for failure: **open inferior capsular shift (Neer)** or **arthroscopic capsular plication ± rotator interval closure**. **Avoid surgery in voluntary/muscle-patterning instability (Polar III)** and **EDS** (high failure).

## 8. Examination tests

- **Anterior: apprehension test** (abduction-external rotation), **relocation test (Jobe: posterior force relieves apprehension)**, **release (surprise) test** (**most specific**), **anterior drawer**, **load and shift**.
- **Posterior: Jerk test**, **Kim test**, posterior drawer, posterior load and shift.
- **Inferior/MDI: sulcus sign**, **Gagey**.
- **Laxity ≠ instability:** instability = symptomatic pathological translation.

Exam pearl: **Age at first dislocation is the strongest predictor of recurrence (under 20: up to 90%). Over 40: think cuff tear and axillary nerve. IGHL anterior band = main restraint in abduction-external rotation.**

Exam pearl: **Glenoid track = 0.83D - d; HSI over the track = off-track (engaging) = add remplissage. Glenoid loss over 20-25% (subcritical 13.5-17%) = Latarjet. ISIS over 6 = Latarjet. Latarjet: sling effect of the conjoint tendon is the main mechanism; musculocutaneous nerve at risk.**

Exam pearl: **Posterior dislocation: seizure/electrocution, locked internal rotation, light bulb sign, reverse Hill-Sachs; McLaughlin transfer for 25-50% defects. HAGL = J sign. ALPSA heals medially (higher recurrence). MDI: rehab first; inferior capsular shift; avoid surgery in muscle patterning.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.